

Network Intrusion Analysis

ROWAN MUHOBERAC + JONAH SCHEPERS

Agenda

- Company Overview
- Business Opportunity
- Financials
- Phishing Modeling
- APT Modeling
- Zero-day Modeling
- ITD Modeling
- Simulator
- Results
- Next Steps

Company Overview

Enterprise Network Security

- Automated cybersecurity services for companies
- Detects 11 of the most common and dangerous network intrusions



Business Opportunity

- Develop specialized models that precisely identify threats

- Specifically looking at Advanced Persistent Threats (APT), Phishing, Zero-day Attacks, and Internal Threat Detection (ITD).

- Success Metrics
  1. Mean Time To Detect (MTTD) decreased by 50%
  2. False Positive Rate <2%
  3. Track Precision and Recall

Financials

Projected Costs

- Total $1.2 million
- Personnel: $500,000
- Infrastructure: $600,000
- Software Licenses: $50,000
- Training: $50,000

Benefits

- Total Benefit: $1.3 million/yr
- Cost Savings: $133,000/yr
- Additional Revenue: $1.166 million/yr

ROI

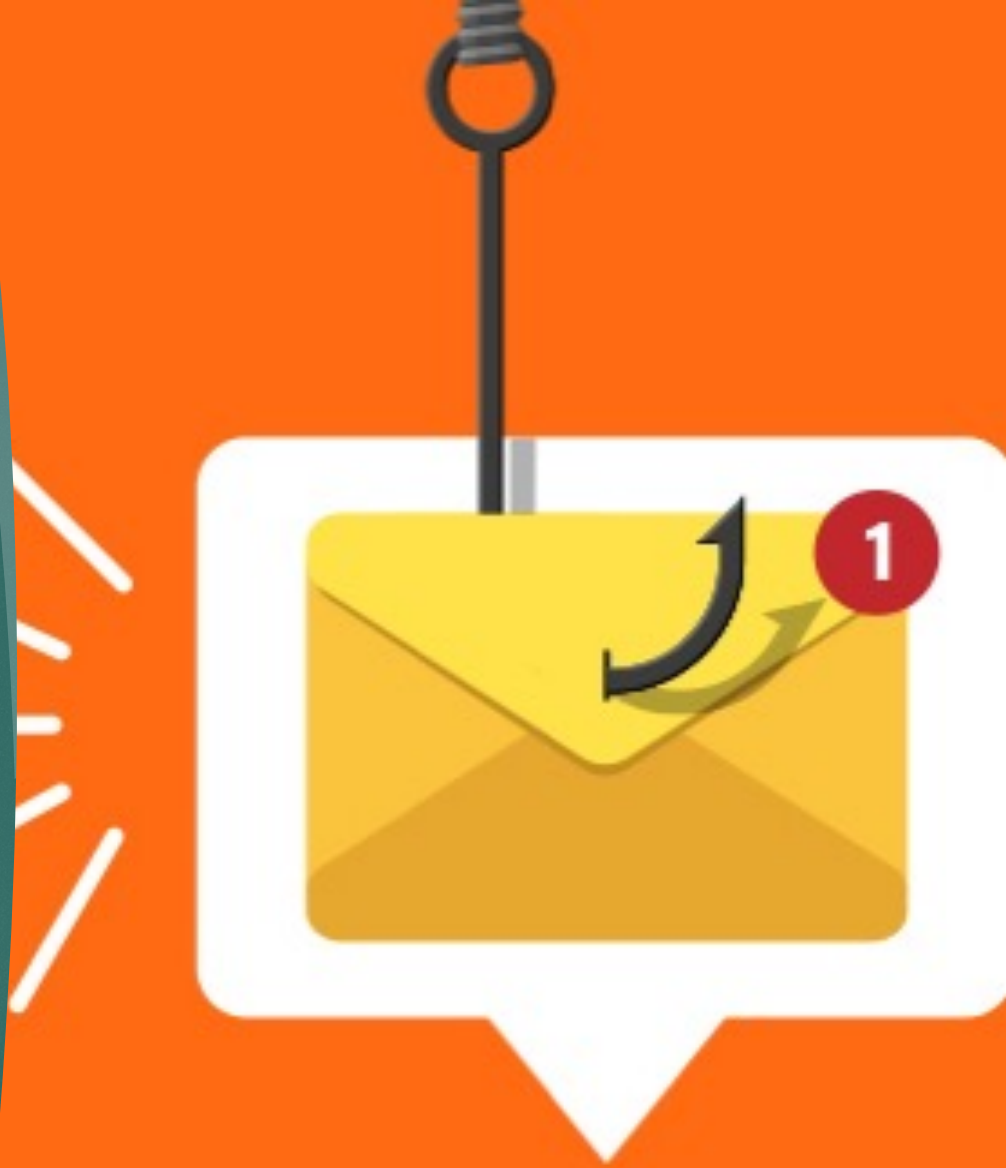
- First year ROI: 7.7% (With Costs)
- Subsequent years ROI: 108%
- Break-even: ~1 Year with Operational costs
- 3-year NPV: $3.152 million (8% discount rate)

Analytics Process



Results for Phishing

- 99.98% of cases correctly identified
- 99.78% Detection - Current System
- 100% of Phishing cases correctly identified
- 0.02% False Positive Rate (<2% Goal)
- Mean Time to Detect 0.6s (0.9s Goal)



Results for APT

- 99.99% of cases correctly identified
- 96% Detection - Current System
- 100% of APT cases correctly identified
- 0.01% False Positive Rate (<2% Goal)
- Mean Time to Detect 0.6s (1s Goal)



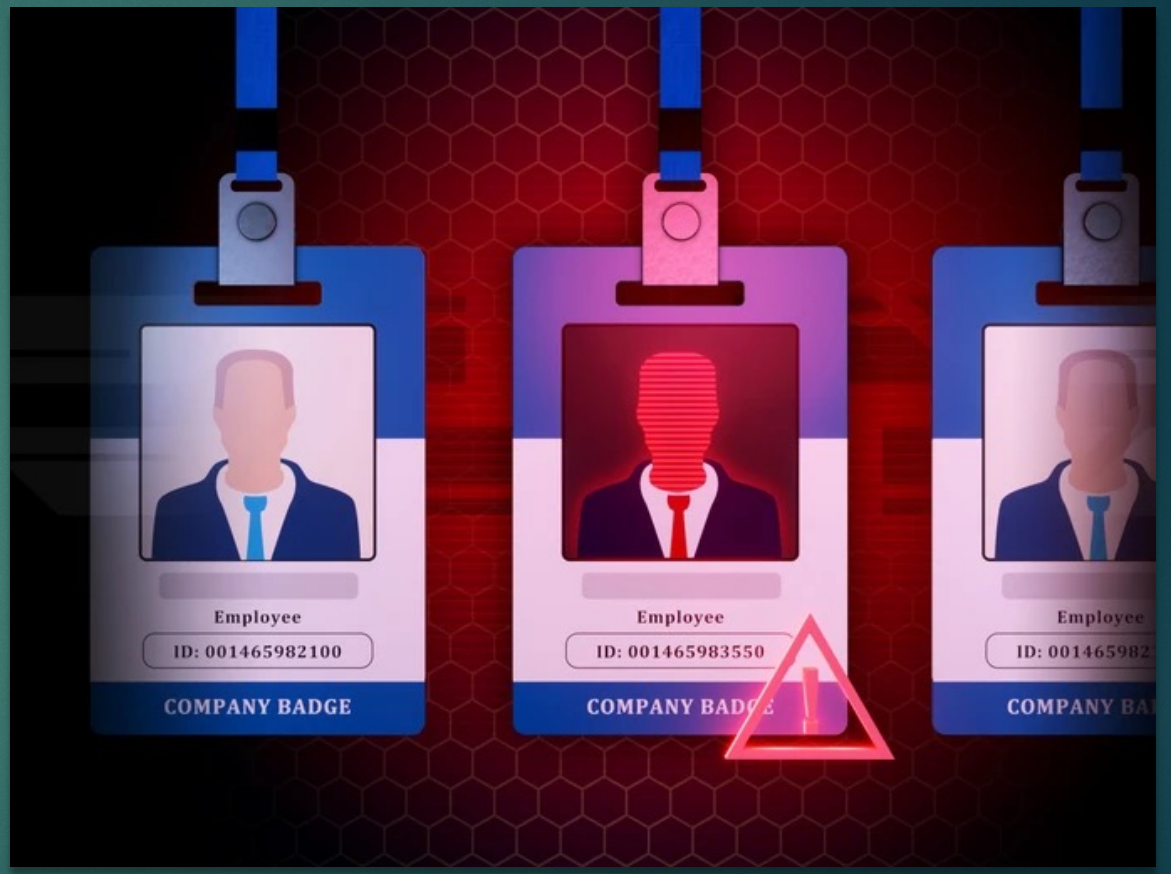
Results for Zero-day

- 100% of cases correctly identified
- 99.8% Detection – Current System
- 100% of Zero-day cases correctly identified
- 0% False Positive Rate (<2% Goal)
- Mean Time To Detect is 0.4s (0.8s Goal)



Results for ITD

- 99.66% of cases correctly identified
- 98.75% Detection – Current System
- 75.61% of ITD cases correctly identified
- 0.2% False Positive Rate (<2% Goal)
- Mean Time To Detect is 0.7s (1.1s Goal)



What Worked



CONNECTION FREQUENCY (ALL MODELS)



INTERNAL IP (ITD)



DURATION (APT)



PACKETS (ZERO-DAY)

What Did Not Work

- Internal and External destination IPs
- Binned Times
- Scaled bytes per packet and packets per second and an interaction of the two
- Off hours

Cybersecurity Simulator

Results (Gain Over Current)

- ~12 more ITD cases detected
  - ~$9.2 million saved
- ~2 more Phishing cases detected
  - ~$8 million saved
- ~2 more Zero-Day case detected
  - ~$4 million saved
- ~27 more APT cases detected
  - ~$100 million saved



Next Steps

- Run A/B Testing against current service to identify actual gains
- Improve models with more network data
- Fully deploy models

Questions?